



FarmerChat Mobile UI Automation

Automated Testing with Maestro Framework

Documentation | April 2026



Overview

I have implemented **automated UI testing** for the FarmerChat Android app using **Maestro**, a mobile UI testing framework. This enables consistent, repeatable testing of critical user journeys without manual intervention.

What is Maestro?

Maestro is an open-source mobile UI testing framework that works on real Android devices and emulators. It uses simple YAML-based test scripts and provides reliable element detection and interaction capabilities.



Quick Stats

5

TEST CASES

100%

CORE FLOWS

~12m

EXECUTION TIME

3x

RETRY LOGIC

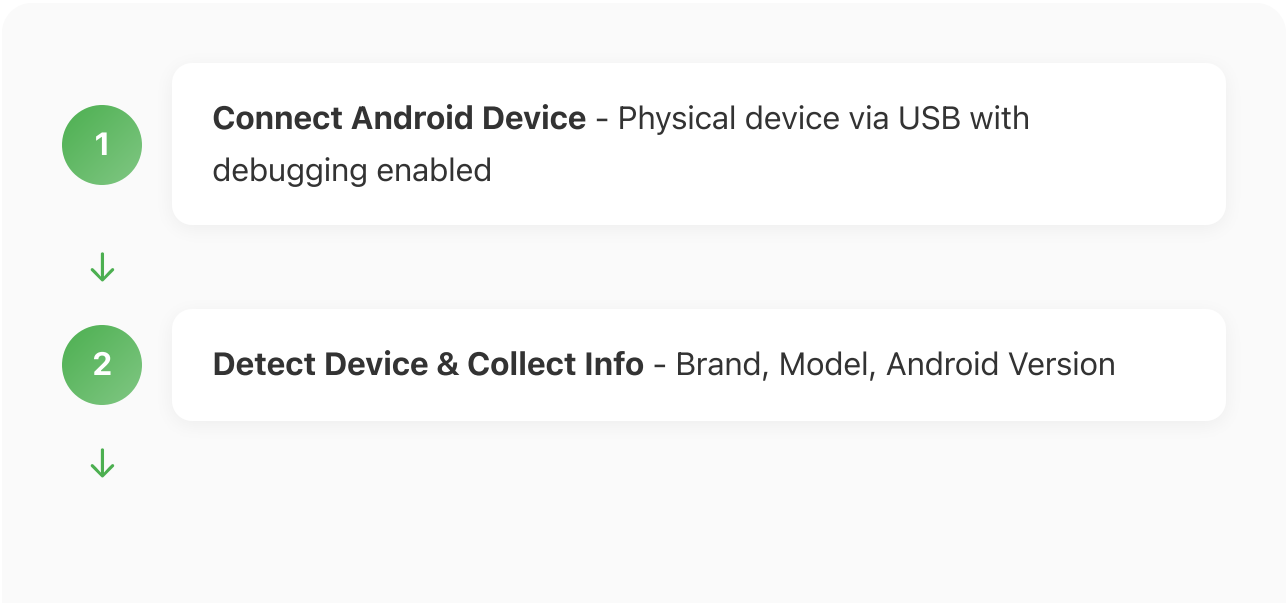


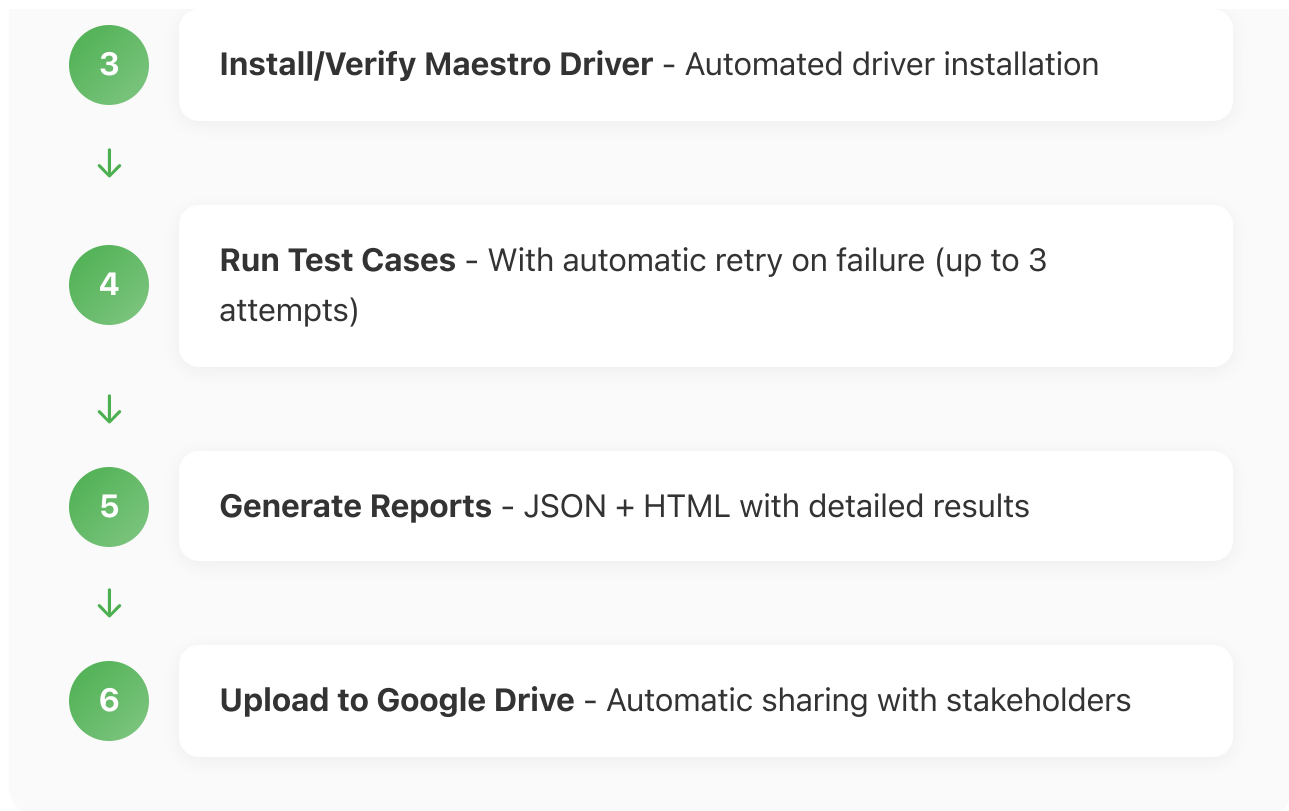
Test Scenarios Automated

ID	TEST CASE	PRIORITY	WHAT IT VALIDATES
TC01	Location-Based Personalization	P0	Ensures the app captures user GPS via the weather widget and displays relevant image questions and content cards based on location
TC02	AI Chat Experience	P0	Validates that users can ask farming-related questions and receive AI-generated responses along with suggested follow-up questions
TC03	Home Feed Usability	P1	Confirms that users can smoothly scroll through the home feed and access all content cards without issues
TC04	Audio Response Feature	P0	Ensures users can listen to AI responses using the text-to-speech feature
TC05	User Authentication & Logout	P0	Verifies complete user flow including sign-up, login, and logout functionality



Test Execution Flow





✨ Key Features



Automatic Retry Logic

Each test retries up to 2 times if it fails. Handles transient issues like network delays and driver timeouts automatically.



Detailed Failure Reporting

Captures error type, failed step, helpful hints, and full log paths for quick debugging.



Comprehensive Logging

Timestamped logs directory, master log file, and individual test attempt logs for complete traceability.



Automated Reports

Machine-readable JSON and stakeholder-friendly HTML reports with pass/fail statistics and device details.



Sample Test Output

FARMERCHAT MAESTRO TEST SUITE

Tester: QA_Team
Device: motorola edge 50 fusion
Android: 16 (SDK 36)

RUNNING TEST CASES

[1/5] Location-Based Personalization	✓ PASSED (59s)
[2/5] AI Chat Experience	✓ PASSED (103s) (retry 1)
[3/5] Home Feed Usability	✓ PASSED (68s)
[4/5] Audio Response Feature	✓ PASSED (73s)
[5/5] User Authentication & Logout	✓ PASSED (69s)

TEST RESULTS SUMMARY

Total: 5
Passed: 5
Failed: 0
Pass Rate: 100%
Duration: 12m 30s

✓ Reports uploaded to Google Drive

Benefits



Consistency



Speed

Same tests run identically every time,
eliminating human error

Full regression in ~12 minutes vs 45+
minutes manual testing



Early Bug Detection

Catch UI regressions before they reach
production



Documentation

HTML reports serve as test evidence for
compliance



Reliability

Retry logic handles flaky scenarios
automatically



Scalability

Easy to add new test cases as features grow

Repository: github.com/Mohamedimran5307/FarmerChat_Core_scenarios

FarmerChat UI Automation Suite • Built with Maestro